

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

POSPrinter (TM-P60/ TM-P60PEELER)

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Section 1. Introduction

This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-P60/ TM-P60PEELER Series POSPrinters are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

Serial	IEEE 802.11b
TM-P60	TM-P60W
TM-P60PEELER	TM-P60PEELERW

Before reading the manual, see the following explanation about the characteristic of the TM-P60/ TM-P60PEELER models.

- Station: Receipt (Line Thermal 203 dpi X 203 dpi)
- Peeler function

Throughout the manual, the various model names will be referred to as TM-P60/ TM-P60PEELER, TM-P60PEELER.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the TM-P60/ TM-P60PEELER Series printers.

2.1 Reference Versions of Firmware

Refer to the release notes (Relnote.txt).

2.2 Settings of DIP Switches

Confirm that the following settings have been made correctly.

1) Serial port

DipSW-1

No	Setting	
1	OFF	Fixed at OFF
2	ON	Fixed at ON
3	OFF	Fixed at OFF
4	ON	Fixed at ON

2) IEEE 802.11b port

Same with the Serial port information.

2.3 Port Information

1) Port information when serial port is used

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Baud rate [bps]	9600, 19200, 38400
Bit length [bit]	8
Parity	NONE, ODD, EVEN
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Baud rate [bps]	19200
Bit length [bit]	8
Parity	NONE
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	1024
Output interval time [ms]	2500

2) Port information when using IEEE 802.11b port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	1024
Output interval time [ms]	2500

2.4 Device Settings

The following explanation is about the settings for TM-P60/ TM-P60PEELER and TM-P60PEELER.

2.4.1 Usable Device Specific Settings

For the TM-P60/ TM-P60PEELER, the following Device Specific Settings are settable by the SetupPOS utility. For more details on each setting, refer to the Section 2 in the Application Development Guide (POSPrinter EPSON TM series).

Tab	Settings
General	Disable panel buttons
	Assume print complete when data output finishes
	Ignore firmware version check
	Homogenize Error Codes
	Output complete timeout
Paper ^{*1}	Paper Width [mm]: LineWidth [dot]: LineCharsList
Bitmap	TMFlogo...
	NVRAM
Color Bitmap	Method
	Brightness
	Primary
Status Log	ERROR
	OFFLINE
	Log file name (full path name)
	Maximum file size [KB]
Special Font ^{*2}	Special Font

^{*1} Available only for the TM-P60.

^{*2} Available only for the TM-P60PEELER.

Section 3. Function Details

This section describes the functions of the TM-P60/ TM-P60PEELER printer in details. Supplementary explanation of the parts not described in detail in the “UPOS” is also given here.

3.1 Property Set Values and Default Values

The following explanation is about the property setting values and the default values.

3.1.1 Capability Set Values

The following values are the Capability set values.

Capability Name	Setting Value
CapTransaction	TRUE
CapCoverSensor	TRUE
CapConcurrentRecSlp	FALSE
CapConcurrentJrnSlp	FALSE
CapConcurrentJrnRec	FALSE
CapConcurrentPageMode	FALSE
CapCharacterSet	PTR_CCS_ASCII
CapMapCharacterSet	FALSE
CapJrnUnderline	FALSE
CapJrnNearEndSensor	FALSE
CapJrnItalic	FALSE
CapJrnEmptySensor	FALSE
CapJrnDwideDhigh	FALSE
CapJrnDwide	FALSE
CapJrnDhigh	FALSE
CapJrnColor	0
CapJrnCartridgeSensor	0
CapJrnBold	FALSE
CapJrn2Color	FALSE
CapJrnPresent	FALSE
CapRecPageMode	TRUE
CapRecUnderline	TRUE
CapRecStamp	FALSE

CapRecRotate180	TRUE
CapRecRight90	TRUE
CapRecPapercut	(TM-P60) TRUE (TM-P60PEELER) FALSE
CapRecNearEndSensor	FALSE
CapRecMarkFeed	(TM-P60) 0 (TM-P60PEELER) *1 0 PTR_MF_TO_TAKEUP PTR_MF_TO_NEXT_TOF PTR_MF_TO_CURRENT_TOF
CapRecLeft90	TRUE
CapRecItalic	FALSE
CapRecEmptySensor	TRUE
CapRecDwideDhigh	TRUE
CapRecDwide	TRUE
CapRecDhigh	TRUE
CapRecColor	PTR_COLOR_PRIMARY
CapRecCartridgeSensor	0
CapRecBold	TRUE
CapRecBitmap	TRUE
CapRecBarCode	TRUE
CapRec2Color	FALSE
CapRecPresent	TRUE
CapSlpUnderline	FALSE
CapSlpRotate180	FALSE
CapSlpRight90	FALSE
CapSlpNearEndSensor	FALSE
CapSlpLeft90	FALSE
CapSlpItalic	FALSE
CapSlpEmptySensor	FALSE
CapSlpDwideDhigh	FALSE
CapSlpDwide	FALSE
CapSlpDhigh	FALSE
CapSlpColor	0
CapSlpCartridgeSensor	0
CapSlpBothSidesPrint	FALSE
CapSlpBold	FALSE
CapSlpBitmap	FALSE
CapSlpBarCode	FALSE

CapSlp2Color	FALSE
CapSlpFullslip	FALSE
CapSlpPresent	FALSE
CapSlpPageMode	FALSE

^{*1} The setting values of CapRecMarkFeed differ depending on the device state and the paper type in the case of the TM-P60PEELER.

Please refer to "[3.17.1 MarkFeed function of TM-P60PEELER](#)" of this manual for details.

3.1.2 List Properties

The List Properties are explained in the following.

List Property	Settings
CharacterSetList	"254 ^{*1} , 255, 437, 737, 850, 851, 852, 853, 857, 858, 860, 863, 865, 866, 998, 999, 1252"
JrnLineCharsList	""
RecLineCharsList (When 57.5 mm is set)	(TM-P60) "35, 42, 52" (TM-P60 Peeler) "32, 38, 48, 16 ^{*1} "
RecLineCharsList (When 59.5 mm is set) ^{*2}	"36, 43, 54"
SlpLineCharsList	""
RecBarcodeRotationList	"0, R90, L90, 180"
RecBitmapRotationList	"0, R90, L90, 180"
SlpBarcodeRotationList	""
SlpBitmapRotationList	""
FontTypefaceList	""

^{*1} When "Special Font" is set. (Available only for the TM-P60PEELER.)

^{*2} The values of RecLineCharsList property are updated by the values specified of "PTR_DI_SET_PAPERLAYOUT" command. (Available only for the TM-P60PEELER.)

3.1.3 Width and Height Properties

The width and height properties are described below.

Property	Settings		
	Default Value	Maximum value [dot]	Minimum value [dot]
RecLineSpacing	30	127	24 ^{*1}
JrnLineSpacing	X	X	X
SlpLineSpacing	X	X	X
SlpLineHeight [dot]	X		
RecLineHeight [dot]	24,16 48 (Special Font) ^{*4}		
JrnLineHeight [dot]	X		
SlpLineWidth [dot]	X		
RecLineWidth [dot] (When 57.5 mm is set)	420 (TM-P60) 388 (TM-P60PEELER)		
RecLineWidth [dot] (When 59.5 mm is set)	432		
JrnLineWidth [dot]	X		
RecSidewaysMaxLines ^{*2}	14		
RecSidewaysMaxChars (When Font A is selected)	100		
RecSidewaysMaxChars (When Font B is selected)	120		
RecSidewaysMaxChars (When Font C is selected)	150		
RecLinesToPaperCut	3 (TM-P60) ^{*3} 4 (TM-P60PEELER) ^{*3}		
SlpSidewaysMaxLines	X		
SlpSidewaysMaxChars	X		
SlpMaxLines	X		

X : No settings

^{*1} In the case of a line thermal station, the Line Spacing setting is identical with the height of the characters which means that it can be set at up to 16 when Font C is selected.

^{*2} It can be changed by the settings of the RecLineSpacing or the RecLineHeight.

^{*3} It can be changed by the settings of the RecLineSpacing or the character height.

^{*4} Available only for the TM-P60PEELER.

3.1.4 Common Property Strings

The Device information properties are described below.

● TM-P60

I/F	DeviceName	DeviceDescription
S	TM-P60	EPSON TM-P60 POS Printer
W	TM-P60W	EPSON TM-P60W POS Printer

● TM-P60PEELER

I/F	DeviceName	DeviceDescription
S	TM-P60PEELER	EPSON TM-P60PEELER POS Printer
W	TM-P60PEELERW	EPSON TM-P60PEELERW POS Printer

I/F shows the connected interface.

There are four types of the connecting interfaces as follows:

S: Serial

W: IEEE 802.11b

3.1.5 PageMode Print Properties

The Device information properties are described below.

Property	Station ^{*2}		
	Journal	Receipt	Slip
PageModeArea	-	(58mm) "420", "1200" (60mm) "432", "1200"	-
PageModeDescriptor ^{*1}	-	BM/BC/BMR/BCR	-

^{*1} Following setting values are used for the PageModeDescriptor property.

BM : Bitmap printing is available.

BC : Barcode printing is available.

BMR : Rotated printing of bitmap is available.

BCR : Rotated printing of barcode is available.

^{*2} If the Station's CapRecPageMode and/or CapSlpPageMode property values are FALSE, the PageModeArea property shall have " " and the PageModeDescriptor property shall have "0" respectively as a setting value.

3.2 Methods

The following explanation is about supported/unsupported Methods, and the detailed information.

Method	Supported/Unsupported	Compatibility with the PageMode printing
PrintNormal	O	O
PrintTwoNormal	X	X
PrintImmediate	O	O ^{*1}
PrintBarCode	O	O ^{*2}
PrintBitmap	O	O ^{*3}
PrintMemoryBitmap	O	O ^{*3}
CutPaper	O (1~100: Cutting with one point of the bottom left corner uncut) X (TM-P60PEELER)	X
MarkFeed	X O (TM-P60PEELER)	X O (TM-P60PEELER)
ChangePrintSide	X	X
ValidateData	O	O
TransactionPrint	O	O
SetLogo	O	O
SetBitmap	O	O
RotatePrint	O	X
EndRemoval	X	X
BeginRemoval	X	X
EndInsertion	X	X
BeginInsertion	X	X
ClearPrintArea	O	O
PageModePrint	O	O

O : Supported

X : Unsupported

^{*1} If the specified Station is ready to print, the printing data shall not be stored in the PageMode printing buffer but, instead, go straight to printing. If the Station is not ready to print, an error is returned.

^{*2} If other than "LEFT" is specified for the printing position of barcode, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

^{*3} If other than "LEFT" is specified for the printing position of bitmap, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

3.3 Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

Escape Sequence	Support	Compatibility with the PageMode printing
#P	0~100 X (TM-P60PEELER)	X
#fP	0~100 X (TM-P60PEELER)	X
#sP	X	X
sL	X	X
#B	O	O
tL	O	O
bL	O	O
#R	O	O
#IF	0~9999	O
#uF Base Pitch [inch]	0~ approx. 50 cm	O
#rF Maximum [inch]	X	X
#fT	X	X
bC	O	O
!bC	O	O
#uC	1~2	O
iC	X	X
liC	X	X
#rC	1	O
rvC	O	O
!rvC	O	O
#sC	X	X
#fC	X	X
#E	0~65535	X
tbC	X	X
!tbC	X	X
tpC	X	X
!tpC	X	X
1C	O	O
2C	O	O
3C	O	O
4C	O	O
#hC	1~8	O
#vC	1~8	O
cA	O	O ^{*1}
rA	O	O ^{*1}

IA	O	O
N	O	O

O : Supported

X : Unsupported

Numbers: Settable range

^{*1} Regardless of the PageModeHorizontalPosition property setting, center or right adjust what is to be printed based on the PageModePrintArea property setting in the horizontal direction.

3.4 Printable Barcode Type

The TM-P60/ TM-P60PEELER and TM-P60PEELER allow the following barcode types.

- Code 128
- Code 128 Parsed
- Code 93
- Codabar
- ITF
- Code 39
- JAN 13 (EAN 13)
- JAN 8 (EAN 8)
- UPC-E
- UPC-A
- PDF417 ^{*1}
- MAXI CODE ^{*1}
- QR CODE ^{*1}
- RSS-14 ^{*1}
- RSS_Expanded ^{*1}
- UCC/EAN128 ^{*1}
- RSS-14 Truncated ^{*1}
- RSS Limited ^{*1}
- RSS-14 Stacked ^{*1}
- RSS-14 Stacked Omidirectional ^{*1}
- RSS Expanded Stacked ^{*1}
- Composite ^{*1}

^{*1} Available only for the TM-P60PEELER.

3.5 MAXI CODE Printing

3.5.1 Symbology Parameter

When printing MAXI CODE, set the Symbology parameter to one of the following values.

PTR_BCS_MAXICODE : Print using MAXI mode 2.

PTR_BCS_OTHER + 0 : Print using MAXI mode 3.

PTR_BCS_OTHER + 1 : Print using MAXI mode 4 or 5. The mode is set to 4 or 5 automatically depending on the length of the Data parameter. (If the data is long, then the data correction level is lowered for printing.)

PTR_BCS_OTHER + 2 : Print using MAXI mode 6.

3.5.2 Printing Size

Because the size of MAXI CODE is fixed, printing is done at a fixed size that is unaffected by the Width and Height parameters. An error occurs only when the Width and Height parameters fall below zero. If the two dimensional barcode cannot fit into the print area (depending on the paper width, layout settings, etc.) then OPOS_E_ILLEGAL is returned and at this moment ResultCodeExtended becomes zero.

3.5.3 Printing Position

Like the one-dimensional barcode, the print position of the two dimensional barcode is the specified position.

3.5.4 Data Format

3.5.4.1. Mode 2

In the case of mode 2, because the format of header + primary message + secondary message is fixed, data for the Data parameter must follow this format.

The header part contains the following data, which can be omitted.

"D> RS 01 GS yy" (In hexadecimal: 0x5B 0x29 0x3E 0x1E 0x30 0x31 0x1D 0x.. 0x..)

yy is '0' to '9' (0x30 to 0x39)

The primary message part contains the following data.

"Postal Code" GS "ISO Country Code" GS "Service Class Code" GS

Field	Length (byte)	Type
Postal Code	1 to 9	Number (0x30 to 0x39)
ISO Country Code	1 to 3	Number (0x30 to 0x39)
Service Class Code	1 to 3	Number (0x30 to 0x39)

For the secondary message, you can freely specify any data from 0x01 to 0xFF, and the data can be omitted. Since the length of the encoded data is not known, the data length cannot be correctly verified. For this reason, if the printer determines, after examining the original data length, that the data can more or less be printed, it tries to print the data; if the length is more than that, then an error occurs. (For the secondary message, an error occurs if the data length is more than 70 bytes.)

3.5.4.2. Mode 3

In the case of mode 3, except for the format of the primary message, it is same as mode 2. The primary message of mode 3 has the following type of data.

"Postal Code" GS "ISO Country Code" GS "Service Class Code" GS

Field	Length (byte)	Type
Postal Code	1 to 6	Number (0x30 to 0x39) A-Z space "\$%&'()*+,-./
ISO Country Code	1 to 3	Number (0x30 to 0x39)
Service Class Code	1 to 3	Number (0x30 to 0x39)

3.5.4.3. Modes 4 and 5

0x01 to 0xFF can be specified to the Data parameter, and there is no restriction on the format. Since the length of the encoded data is not known, the data length cannot be correctly verified. For this reason, if the printer determines, after examining the original data length, that the data can more or less be printed, it tries to print the data; if the length is more than that, then an error occurs. (An error occurs if Data is zero byte or more than 80 bytes.)

Mode 4 differs from mode 5 in the error correction level. As much as possible, OPOS uses the higher error correction level. Therefore, if the data amount is small (less than 50 bytes), OPOS uses mode 5 (the one with a higher error correction level).

3.5.4.4. Mode 6

0x01 to 0xFF can be specified to the Data parameter, and there is no restriction on the format. Since the length of the encoded data is not known, the data length cannot be correctly verified. For this reason, if the printer determines, after examining the original data length, that the data can more or less be printed, it tries to print the data; if the length is more than that, then an error occurs. (An error occurs if Data is zero byte or more than 80 bytes.)

3.6 QR CODE Printing

3.6.1 Symbology Parameter

When printing QR CODE, set the Symbology parameter to one of the following value.

PTR_BCS_OTHER + 3 : Print using QR CODE model 1 (old specification, used for maintaining compatibility).

PTR_BCS_OTHER + 4 : Print using QR CODE model 2.

3.6.2 Printing Size

Because the width and length of QR CODE are the same, printing is done to the inner part at a size closest to it by using the value specified by the Width parameter. Therefore, the height of print is not affected by the Height parameter. If the Height parameter is less than 0, an error occurs.

The print size is determined by the version of QR and the size of the module. Because the version of QR is determined by the data length and type, you can use the size of the module to adjust the print size. If the two dimensional barcode cannot fit into the print area (depending on the paper width, layout settings, etc.) then OPOS_E_ILLEGAL is returned and at this moment ResultCodeExtended becomes zero.

For QR, it differs from other two dimensional barcodes; if the encoded data result is not known, then the print width cannot be obtained. If the print width cannot be obtained, the page mode range for 90-degree rotated printing cannot be specified. Therefore, within OPOS it calculates the number of code words of the encoded data. Because of this reason, data amount can be correctly verified.

3.6.3 Error Correction Level

Error correction level is fixed at 7%.

3.6.4 Printing Position

Like the one dimensional barcode, the print position of the two dimensional barcode is the specified position.

3.7 RSS Printing

3.7.1 Symbology Parameter

When printing RSS, set the Symbology parameter to one of the following value.

- PTR_BCS_OTHER + 7 : Print using RSS 14 Stacked.
- PTR_BCS_OTHER + 8 : Print using RSS 14 Stacked Omnidirectional.
- PTR_BCS_OTHER + 8 : Print using RSS Expanded Stacked.

3.7.2 Printing Size

Printing is done to the inner part at a size closest to it by using the value specified by the Width parameter.

Therefore, the height of print is not affected by the Height parameter. If the Width and Height parameters are less than 0, an error occurs.

Because the printing size is determined by the data length ^{*1} and barcode type, you can use the size of the module to adjust the print size.

If the two dimensional barcode cannot fit into the print area (depending on the paper width, layout settings, etc.) then OPOS_E_ILLEGAL is returned and at this moment ResultCodeExtended becomes 0.

^{*1} Available only for the RSS Expanded Stacked.

3.7.3 Printing Position

Like the one dimensional barcode, the print position of the two dimensional barcode is the specified position.

3.7.4 Data Format

[Setting range of data]

Symbology	Length (byte)	Characters that can be specified
RSS 14 Stacked	13	0x30-0x39
RSS 14 Stacked Omnidirectional	13	0x30-0x39 (The first character is limited to 0x30 or 0x31.)
RSS Expanded Stacked	2 to 255	0x20-0x22, 0x25-0x3f, 0x41-0x5a, 0x5F, 0x61-0x7a (The first two characters are limited to 0x30-0x39. Or if the first character is 0x28, the second and the third characters are limited to 0x30-0x39.)

[Special characters of EAN 128]

Special characters	ASCII
FNC1	{1
'('	{{
')'	}

3.8 COMPOSITE Printing

3.8.1 Symbology Parameter

When printing COMPOSITE, set the Symbology parameter to the following value.

HIWORD : The constant value of PDF417

LOWORD : The constant value of the barcode that will combine the value of PDF417.

3.8.1.1. Combinable Barcode

The settable barcode in the ROWORD is as follows.

- UPC-A
- UPC-E (Compressed format)
- UPC-E
- EAN 8
- EAN 13
- RSS-14
- RSS-14 Truncated
- RSS-14 Stacked
- RSS-14 Stacked Omnidirectional
- RSS Limited
- RSS Expanded
- RSS Expanded Stacked
- UCC/EAN128

3.8.2 Printing Size

Printing is done to the inner part at a size closest to it by using the value specified by the Width parameter.

Therefore, the height of print is not affected by the Height parameter. If the Width and Height parameters are less than 0, an error occurs.

Because the printing size is determined by the data length and composite barcode type, you can use the size of the module to adjust the print size.

If the two dimensional barcode cannot fit into the print area (depending on the paper width, layout settings, etc.) then OPOS_E_ILLEGAL is returned and at this moment ResultCodeExtended becomes 0.

3.8.3 Printing Position

Like the one dimensional barcode, the print position of the two dimensional barcode is the specified position.

3.8.4 Data Format

The range designation of the HIWORD data is as follows.

Symbology	Length (byte)	Characters that can be specified
PDF417	3 to 2361	0x20-0x22, 0x25-0x3f, 0x41-0x5a, 0x5f, 0x61-0x7a

The range designation of the ROWORD data is as follows.

Symbology	Length (byte)	Characters that can be specified
UPC-A	11	0x30-0x39 (The first character is limited to 0x30.)
UPC-E (Compressed format)	6	0x30-0x39
UPC-E	11	0x30-0x39 (The first character is limited to 0x30.)
EAN 8	7	0x30-0x39
EAN 13	12	0x30-0x39
RSS-14	13	0x30-0x39
RSS-14 Truncated	13	0x30-0x39
RSS-14 Stacked	13	0x30-0x39
RSS-14 Stacked Omnidirectional	13	0x30-0x39 (The first character is limited to 0x30 or 0x31.)
RSS Limited	13	0x30-0x39 (The first character is limited to 0x30 or 0x31.)
RSS Expanded	2 to 255	0x20-0x22, 0x25-0x3f, 0x41-0x5a, 0x5f, 0x61-0x7a (The first two characters are limited to 0x30-0x39. Or if the first character is 0x28, the second and the third characters are limited to 0x30-0x39.)
RSS Expanded Stacked	2 to 255	0x20-0x22, 0x25-0x3f, 0x41-0x5a, 0x5f, 0x61-0x7a (The first two characters are limited to 0x30-0x39. Or if the first character is 0x28, the second and the third characters are limited to 0x30-0x39.)
UCC/EAN128	3 to 255	0x00-0x7f

[Special characters of Barcodes (EAN 128, RSS Expanded, and RSS Expanded Stacked)]

Special characters	ASCII
FNC1	{1
FNC3	{3
{'	{{
{('	{{(
{')'	{{)}
{*'	{{*

3.9 Power Conditions Report

The TM-P60 and TM-P60PEELER supports Power Conditions Report as follows.

Powered on reporting: Supported (Available by the manufacturer settings.)

Powered off reporting: Supported (Available by the manufacturer settings.)

The manufacturer settings are automatically set.

3.10 Synchronous Processing

The TM-P60/ TM-P60PEELER and TM-P60PEELER support the Process ID for the Synchronous Processing.

Use of the Process ID allows multiple print commands to be queued to the printer simultaneously. For this reason, Asynchronous output (AsyncMode = TRUE) gives a performance improvement.

3.11 Print Position

The TM-P60/ TM-P60PEELER supports the function for setting printing position.

Function	Support
Left margin	O
Print position	O

O: Supported

X: Unsupported

When supports the left margin setting function, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When supports the printing position settings, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

3.12 Electronic Logo Function (NVRAM)

The TM-P60 models feature an electronic logo function (NVRAM). There are two ways of using NVRAM explained as follows:

1. Using TMFlogo Utility

Start up TMFlogo utility from “Device Specific Settings” dialog box of SetupPOS utility, and register image files (BMP style) with NVRAM in advance. For the details of the registration, please refer to the “Help” of “TMFlogo utility” and/ or “EPSON OPOS ADK MANUAL User’s Manual TMFlogo Utility”.

To print image files registered with NVRAM, please use the following DirectIO:

PTR_DI_FLASH_BITMAP2

Please refer to the Section 4 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)” for detail.

2. Using SetBitmap Method

Checking “NVRAM” check box on “Bitmap” tab in “Device Specific Settings” dialog box of SetupPOS utility enable to register image files with NVRAM using SetBitmap method.

Regarding the details of image files registration with NVRAM using SetBitmap method, please refer to the Section 8 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”

The available NVRAM size for the TM-P60/ TM-P60PEELER is 393216 bytes.

The available NVRAM size for the TM-P60/ TM-P60PEELERPEELER is 393216 bytes.

3.13 Printable bitmap types and the specified size

The TM-P60 models support the following bitmap commands. For the detailed information about the each command, refer to the Section 2 in the Application Development guide (POSPrinter EPSON TM Series). The allowance ranges for bitmaps are as follows.

Bitmap command type	Allowance range		
	X	Y	xy
NV bitmap	1~1000	1~1200	
Raster bitmap	1~1000	1~1200	

3.14 Maintenance Counter

The TM-P60/ TM-P60PEELER models feature a maintenance counter function for retaining an operation log of the printer. The following chart shows the available maintenance counters for the TM-P60/ TM-P60PEELER.

Counter number Hexadecimal	Counter	Unit	Max Value	Counter Type
14	Number of line feeds	Lines	143,165,576	Resettable
15	Number of head energizations: Roll paper	Times	4,294,967,295	Resettable
32 ^{*1}	Number of auto-cutter operations	Times	4,294,967,295	Resettable
46	Uptime of product	Hours	71,582,788	Resettable
94	Number of line feeds	Lines	143,165,576	Cumulative
95	Number of head energizations: Roll paper	Times	4,294,967,295	Cumulative
B2 ^{*1}	Number of auto-cutter operations	Times	4,294,967,295	Cumulative
C6	Uptime of product	Hours	71,582,788	Cumulative

^{*1} Available only for the TM-P60.

3.15 Automatic Recovery Function

The TM-P60 models feature a function for automatic recovery when the power is turned on again after an interruption of power. Recovery processing is performed automatically when the printer's power is turned on again after an interruption. The recovery processing restores the printer to the condition it was in before the power was turned off.

3.16 Output without flow control on the IEEE 802.11b interfaces

The TM-P60/ TM-P60PEELER models support outputting without flow control on the IEEE 802.11b interfaces.

3.17 MarkFeed Function

The TM-P60PEELER support the MarkFeed function.

The OPOS_E_ILLEGAL / OPOS_EX_INVALIDMODE is returned at the execution of the PTR_MF_TO_CURRENT_TOF under the condition that the label is at the peel-off position or the position other than cut position in the case of the TM-P60PEELER.

However, the OutputCompleteEvent is fired upon the processing of the transferred command during the asynchronous print since the label position cannot be comprehended.

3.17.1 MarkFeed function of TM-P60PEELER

The operations of the CapRecMarkFeed and the MarkFeed method in the TM-P60PEELER is as follows.

Device Mode	Paper Type	CapRecMarkFeed
Peel-off Issuance Mode	Roll paper	0
	Black marked roll paper	PTR_MF_TO_NEXT_TOF PTR_MF_TO_CURRENT_TOF
	Label paper	PTR_MF_TO_TAKEUP
	Black marked label paper	PTR_MF_TO_CURRENT_TOF
Serial Issuance Mode	Roll paper	0
	Black marked roll paper	PTR_MF_TO_NEXT_TOF PTR_MF_TO_CURRENT_TOF
	Label paper	PTR_MF_TO_TAKEUP
	Black marked label paper	PTR_MF_TO_NEXT_TOF PTR_MF_TO_CURRENT_TOF

3.18 Label Receipt DirectIO

The TM-P60PEELER support the following DirectIO commands. For usage information, please refer to the Section 4 of "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)".

PTR_DI_LABEL_REMOVE

3.19 Label Receipt Usage

- To use the function related to the label, the paper should be label paper or black marked label paper.
- When using label paper, the Escape Sequence ESC|#IF feeds the receipt the specified length, or to the head of the next label.

Section 4. Warnings

This section describes precautions in use of TM-P60/ TM-P60PEELER and TM-P60PEELER.

When using the TM-P60PEELER, if a mechanical-error is returned, it means that there is a layout-error. To avoid the error, set the value of "sa" of PTR_DI_SET_PAPERLAYOUT of the DirectIO command to 0. Or reset the value of the paper layout to an appropriate number.